A Data-Driven Analysis of Phishing Website Characteristics Using Python

1. Overview

Phishing is a common cybersecurity threat in which malicious websites attempt to deceive users and collect sensitive information. This project focuses on analyzing phishing and legitimate websites to identify patterns and characteristics associated with suspicious websites.

The project uses the PhiUSIIL Phishing URL Dataset obtained from the UCI Machine Learning Repository. The original dataset contains 235,795 records, and a sample of 49,999 records with 56 columns was used for this analysis.

Python was used to perform data cleaning, exploratory data analysis (EDA), feature engineering, statistical analysis, and data visualization. The analysis focuses on URL length, domain length, special characters, subdomains, URL similarity scores, HTTPS usage, and website features such as titles, favicons, and descriptions.

A Logistic Regression model was also developed to classify phishing and legitimate websites based on their characteristics.

2. Objectives of the Study

- Analyze URL and domain length patterns between phishing and legitimate websites.
- Examine special characters, digit usage, and URL obfuscation.
- Perform statistical analysis of subdomain counts.
- Compare URL similarity scores and domain-title matching scores.
- Analyze website features such as titles, favicons, descriptions, and password fields.
- Examine HTTPS usage, URL redirects, and other website security indicators.
- Perform exploratory data analysis and identify relationships between website characteristics.
- Create meaningful visualizations and interpret the findings.
- Develop and evaluate a Logistic Regression classification model.
- Derive useful insights and recommendations based on the analysis.

3. Tools and Technologies Used

Python: Primary programming language for data analysis and predictive modeling.

Pandas: Loading, cleaning, organizing, and analyzing the dataset.

NumPy: Numerical calculations and statistical operations.

Matplotlib: Graphs and visualizations.

Seaborn: Statistical visualizations and correlation analysis.

Scikit-learn: Data scaling, Logistic Regression classification, and model evaluation.

Google Colab: Cloud-based environment for writing and executing Python code.

4. Conclusion

This project analyzed phishing and legitimate websites using Python to identify differences in URL structures, domain characteristics, and website features.

The analysis included 49,999 website records: 20,625 phishing websites and 29,374 legitimate websites. Data validation found no missing values or duplicate records.

Phishing websites had longer average URLs (42.14) than legitimate websites (26.28), and more special characters on average (3.43 versus 1.24). Legitimate websites showed higher domain-title matching scores and a greater presence of titles, favicons, and descriptions.

Subdomain counts differed only slightly between phishing and legitimate websites, suggesting this feature alone is not a strong indicator. HTTPS usage also cannot independently establish website legitimacy.

A Logistic Regression model achieved 99.98% accuracy on the test dataset. After excluding four selected features for additional validation, accuracy was 99.92%. Independent external testing is needed to assess real-world reliability.

Overall, the project demonstrates how data preparation, statistical analysis, visualization, and machine learning can help identify suspicious website characteristics. Multiple features should be evaluated together rather than relying on a single indicator.

Dataset Source: UCI Machine Learning Repository — PhiUSIIL Phishing URL Dataset

https://archive.ics.uci.edu/dataset/967/phiusiil+phishing+url+dataset